

UNIT V: React (Most Scoring Unit)

- **Q1 (The "Leak" Question):** Explain the **React Component Lifecycle Methods** (`componentDidMount`, `componentDidUpdate`, `componentWillUnmount`).
- **Q2:** What is the **Virtual DOM**? How does it make React faster than the Real DOM?
- **Q3:** Explain **State and Props** in React with a code example.
- **Q4:** How do you handle **Forms and Routing** in React?

* The Virtual DOM

DOM

- DOM stands for Document Object Model.
- It is a tree-like structure of your web page.
- Every HTML element is a node in the tree.

Virtual DOM (V-DOM)

- Virtual DOM is a copy of the real DOM.
- It is stored in memory by React.
- React uses Virtual DOM to make changes faster & smoother.
- React use V-DOM to improve performance.

Why V-DOM?

- Changing the real DOM directly is slow.
- ⇒ React updates V-DOM first.
 - whenever input changes
 - React updates VDOM, not the real DOM directly.
- ⇒ It compares old & new Virtual DOM (diffing) → process
 - It checks what has changed?
- ⇒ Only the changed part is updated in the real DOM.
 - React updates only the specific part not the whole page.
- ⇒ No full page reloads.
 - Only changed parts update instantly.

- It saves times & memory.
- Better Uses Experience.
- You don't need to create or manage DOM yourself.
- React handles it automatically.

Example

Let's say you have :

`<h1> hello Meghana </h1>`

then you update it to :

`<h1> hello World </h1>`

⇒ React won't reload the whole Page.

⇒ It only changes specific part
Meghana → World

Benefits of Virtual DOM

- Faster Performance
- Saves time
- Saves Memory
- Smart updates (no full page reload)
- Smooth uses experience
- Useful in large apps like Facebook, Instagram

Realworld examples

- Facebook (only new posts or like updates)
- Amazon (only price or stock updates)
- Gmail (only new email shows up)
- WhatsApp Web

* Rendering & Life Cycle methods in React

Rendering

- Rendering means displaying components on the screen.
- It means generating UI & displaying the UI.
- React components render based on the data.
- When data changes, React automatically re-renders the component.
- Rendering is fast
 - due to VDOM.

LifeCycle Methods

- There are mainly 3 Phases
- Mounting Phase
 - Updating Phase
 - Unmounting Phase
- ⇒ Lifecycle methods are special functions
- ⇒ Only used in class components.

1. Mounting Phase (Component Creation)

- It is the initial phase.
- It is called when a component is created & inserted into DOM.

methods involved;

- constructor()
- render()
- ComponentDidMount()

⇒ these methods run

- when the component is created & added to the DOM

2. Updating Phase (when state or props change)

→ This phase occurs when:

- State changes
- Props changes.

methods involved

- render()
- componentDidUpdate()
- shouldComponentUpdate()

3. Unmounting Phase (Component is Removed)

→ This phase occurs when:
- the component is removed from DOM.

→ only one method runs:
- `Component.prototype.unmount()`

→ Best for:
- Cleanup tasks
- Cancelling API requests
- Removing event listeners

Ex: ①. Like a Person's life:

Mounting → Born

Updating → Grows & Changes

Unmounting → Leaves

②. Instagram

Mounting - Uses taps a Post

Updating - Uses like, comment

Unmounting - Uses swipes down to close Post view

* Working with form in react

- A form is used to take input from user
Ex: Login form, Signup form
- React controls the form using state.
- It updates the state whenever the user types.
- This is called a controlled component
- You use useState to store & update values in a form.

Ex:

Meesho Signup — Name, email, Password

Login Page — Email and Password

Checkout Page — Address, Phone number

Search Box — what user is searching.

* Routing in React

→ Routing in React means navigating b/w Pages
- without reloading the website.

→ It means moving b/w Pages in your app.

Ex: From Home → Cart → login

→ React doesn't have built-in routing

→ So we use a third-party library called React Router

→ React Router build Single-Page Applications (SPA)

→ It loads only the part of the page that changes

Steps to use Routing:

Step-1: Install React Router

```
npm install react-router-dom
```

Step-2: Import it

```
import { BrowserRouter, Routes, Route, Link } from  
    'react-router-dom'
```

Step-3: wrap your app in `<BrowserRouter>`

Step-4: Add routes inside `<Routes>`

```
ex: <Routes>  
  <Route path="/"  
    element={<Home />} />  
  <Route path="/cart"  
    element={<cart />} />  
</Routes>
```

Step-5: use `<Link>` to switch pages

```
<Link to="/cart">Cart </Link>
```

Ex:

- You click on a Product → goes to /product
- You click on cart icon → goes to /cart
- You login → goes to /login